

- 1(a) The pre-release material contains information concerning median house prices over the period 2004 – 2015. A spreadsheet has been used to generate a time series graph for two areas: the London borough of “Barking and Dagenham” and “North West”. This is shown together with the raw data in Fig. 11.1.

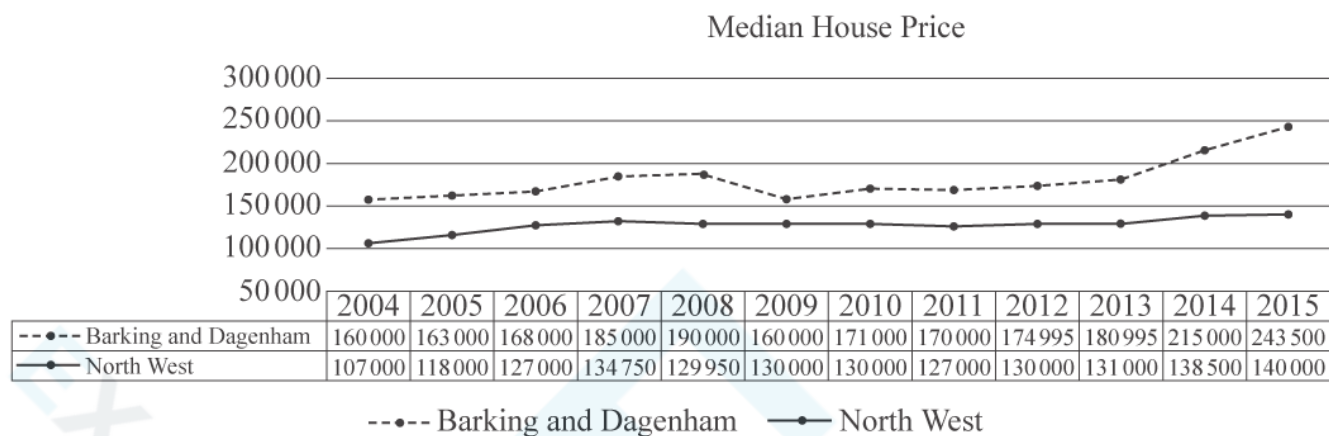


Fig. 11.1

Dr Procter suggests that it is unusual for median house prices in a London borough to be consistently higher than those in other parts of the country.

Use your knowledge of the large data set to comment on Dr Procter's suggestion.

[1]

- (b) Dr Procter wishes to predict the median house price in Barking and Dagenham in 2016. She uses the spreadsheet function LINEST to find the equation of the line of best fit for the given data. She obtains the equation

$P = 4897Y - 9\,657\,847$, where P is the median house price in pounds and Y is the calendar year, for example 2015.

Use Dr Procter's equation to predict the median house price in Barking and Dagenham in

- 2016
- 2017.

[2]



- (c) Professor Jackson uses a simpler model by using the data from 2014 and 2015 only to form a straight-line model.

Find the equation Professor Jackson uses in her model.

[2]

- (d) Use Professor Jackson's equation to predict the median house price in Barking and Dagenham in

- 2016
- 2017.

[2]

- (e) Professor Jackson carries out some research online. She finds some information about median house prices in Barking and Dagenham, which is shown in Fig. 11.2.

2016	2017
£290 000	£300 000

Fig. 11.2

Comment on how well

- Dr Procter's model fits the data,
- Professor Jackson's model fits the data.

[2]

- (f) Explain which, if any, of the models is likely to be more reliable for predicting median house prices in Barking and Dagenham in 2020.

[1]

2 In this question you must show detailed reasoning.

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A circle has centre $(2, -1)$ and radius 5.

A straight line passes through the points $(1, 1)$ and $(9, 5)$.

Find the coordinates of the points of intersection of the line and the circle.

[8]



3(a) (i) Write down the derivative of e^{kx} , where k is a constant.

[1]

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- (ii) A business has been running since 2009. They sell maths revision resources online.
Give a reason why an exponential growth model might be suitable for the annual profits for the business.

[1]

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- (b) Fig. 6 shows the relationship between the annual profits of the business in thousands of pounds (y) and the time in years after 2009 (x). The graph of $\ln y$ plotted against x is approximately a straight line.

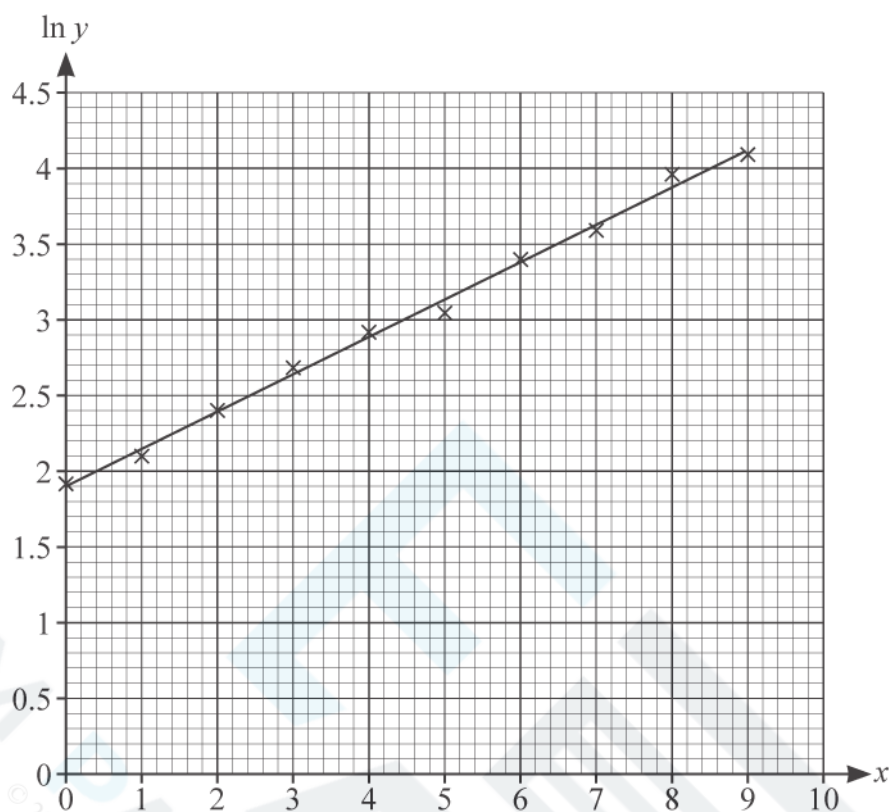
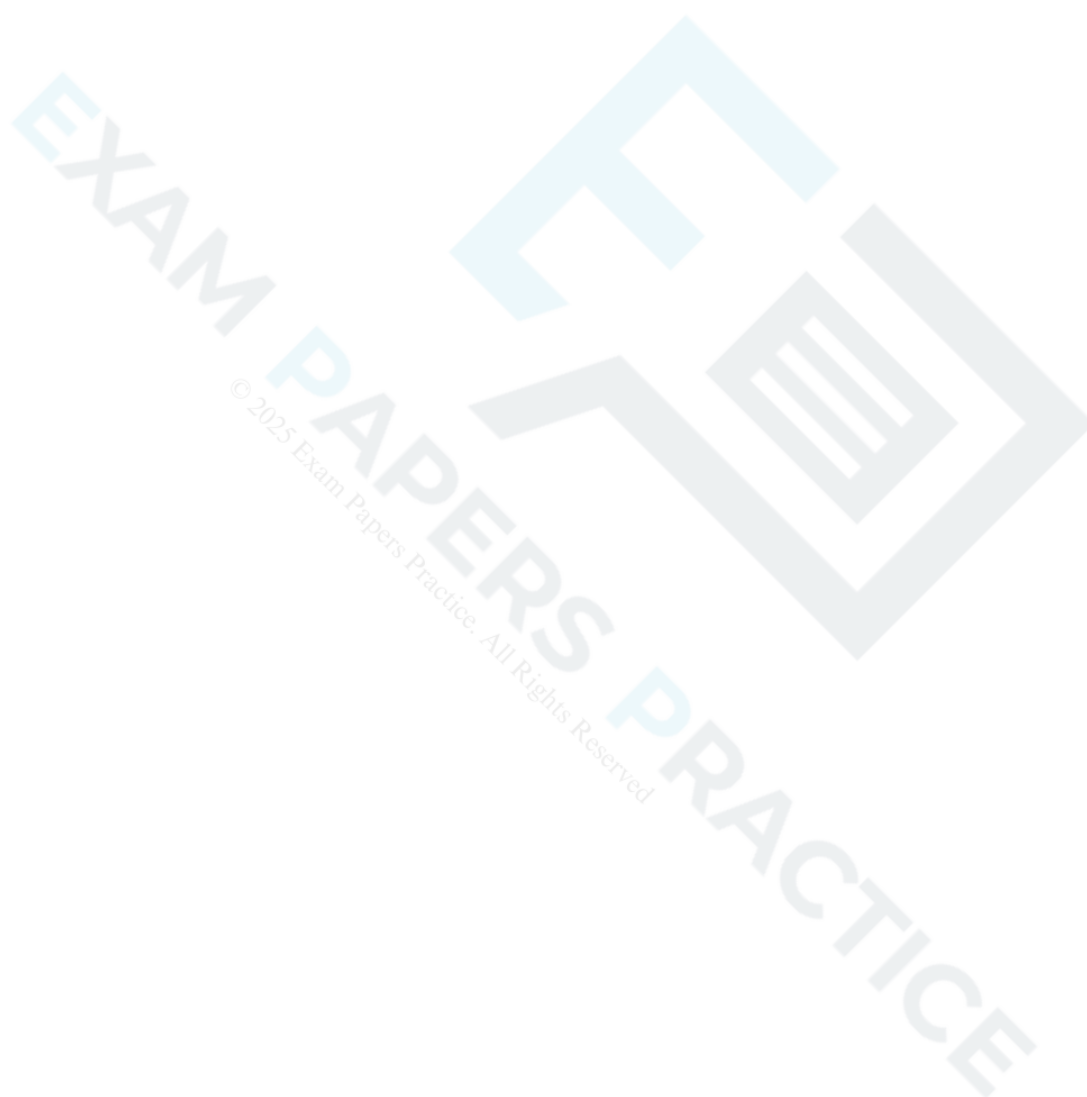


Fig. 6

Show that the straight line is consistent with a model of the form $y = Ae^{kx}$, where A and k are constants.

[2]

(c) Estimate the values of A and k .

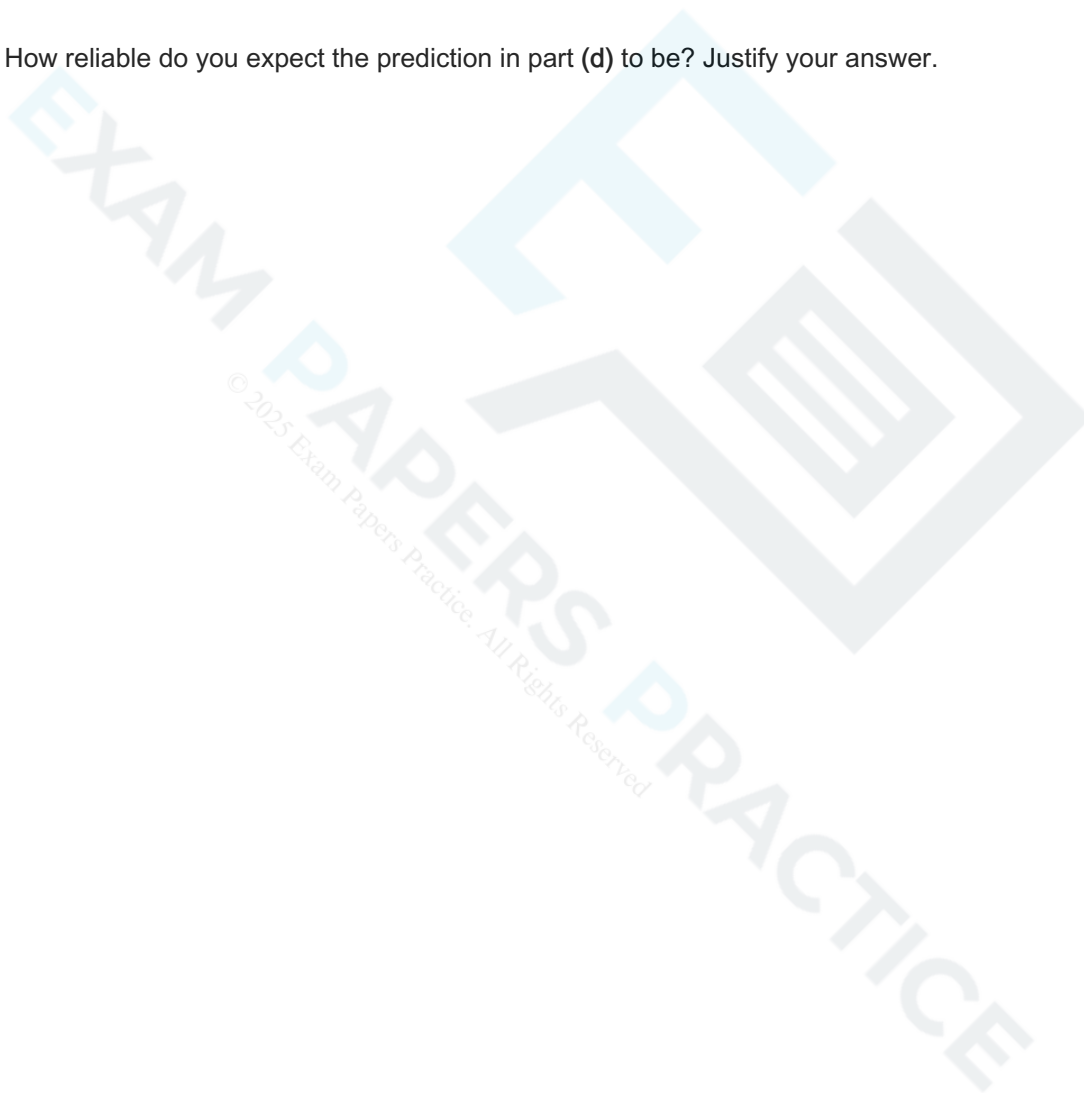


(d) Use the model to predict the profit in the year 2020.

[3]

(e) How reliable do you expect the prediction in part (d) to be? Justify your answer.

[1]



- 4 The point A has coordinates $(-1, -2)$ and the point B has coordinates $(7, 4)$. The perpendicular bisector of AB intersects the line $y + 2x = k$ at P.

Determine the coordinates of P in terms of k .

[7]

- 5 A projectile is fired from ground level at 35 m s^{-1} at an angle of θ° above the horizontal.

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Find the cartesian equation of the trajectory of the projectile.

[4]



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6(a) In this question you must show detailed reasoning.

Fig. 10 shows the curve given parametrically by the equations $x = \frac{1}{t^2}$, $y = \frac{1}{t^3} - \frac{1}{t}$, for $t > 0$.

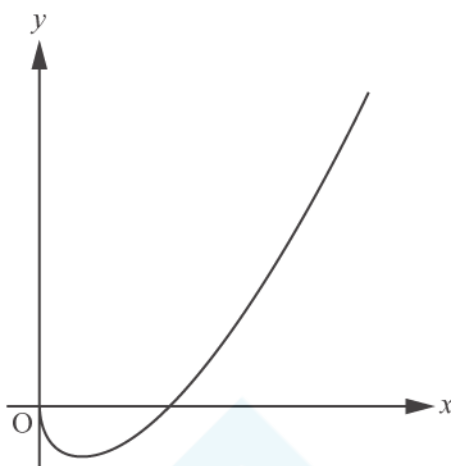


Fig. 10

Show that $\frac{dy}{dx} = \frac{3-t^2}{2t}$.

[3]

- (b) Find the coordinates of the point on the curve at which the tangent to the curve is parallel to the line $4y + x = 1$. [3]

- (c) Find the cartesian equation of the curve. Give your answer in factorised form. [3]

- 7(a) In 2012 Adam bought a second hand car for £8500. Each year Adam has his car valued. He believes that there is a non-linear relationship between t , the time in years since he bought the car, and V , the value of the car in pounds. Fig. 9.1 shows successive values of V and $\log_{10} V$.

t	0	1	2	3	4
V	8500	6970	5720	4690	3840
$\log_{10} V$	3.93	3.84	3.76	3.67	3.58

Fig. 9.1

Adam uses a spreadsheet to plot the points $(t, \log_{10} V)$ shown in Fig. 9.1, and then generates a line of best fit for these points. The line passes through the points $(0, 3.93)$ and $(4, 3.58)$. A copy of his graph is shown in Fig. 9.2.

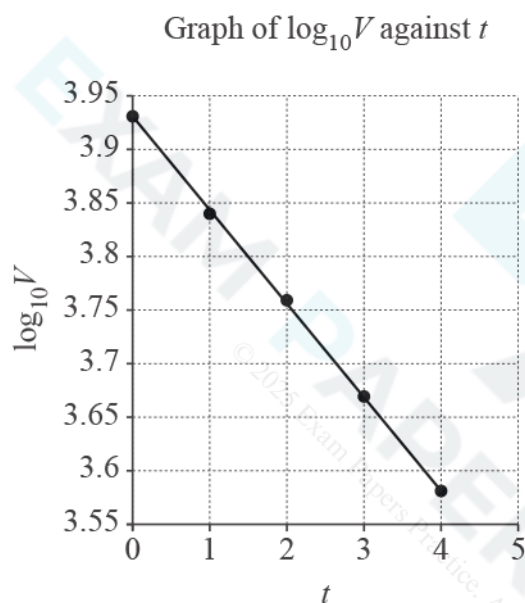


Fig. 9.2

Find an expression for $\log_{10} V$ in terms of t .

[3]



- (b) Find a model for V in the form $V = A \times b^t$, where A and b are constants to be determined. Give the values of A and b correct to 2 significant figures. [3]

- (c) In 2017 Adam's car was valued at £3150.

Determine whether the model is a good fit for this data.

[1]

- (d) A company called Webuyoldcars pays £500 for any second hand car. Adam decides that he will sell his car to this company when the annual valuation of his car is less than £500.

According to the model, after how many years will Adam sell his car to Webuyoldcars?

[3]

- 8 To answer this question, please refer to the Insert for H640-03 June 2019.
Show that the equation of the line in Fig. C2 is $ry+hx=hr$, as given in line 24.

[2]

- 9 In this question you must show detailed reasoning.

The curve $xy+y^2=8$ is shown in Fig. 9.

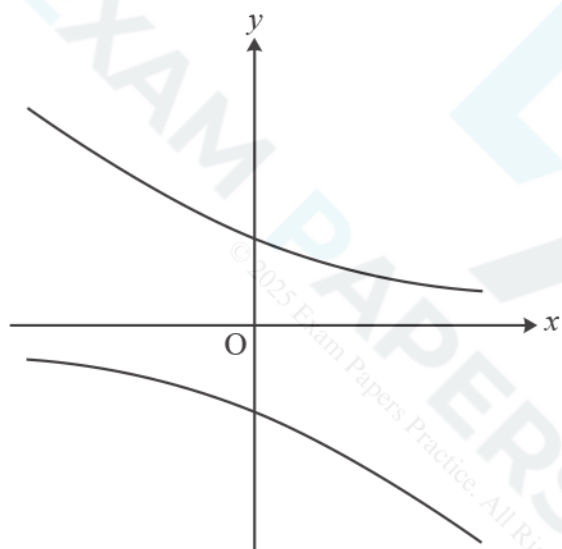


Fig. 9

Find the coordinates of the points on the curve at which the normal has gradient 2.

[6]

10 In this question you must show detailed reasoning.

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A is the point (1, 0), B is the point (1, 1) and D is the point where the tangent to the curve $y = x^3$ at B crosses the x -axis, as shown in Fig. 8. The tangent meets the y -axis at E.

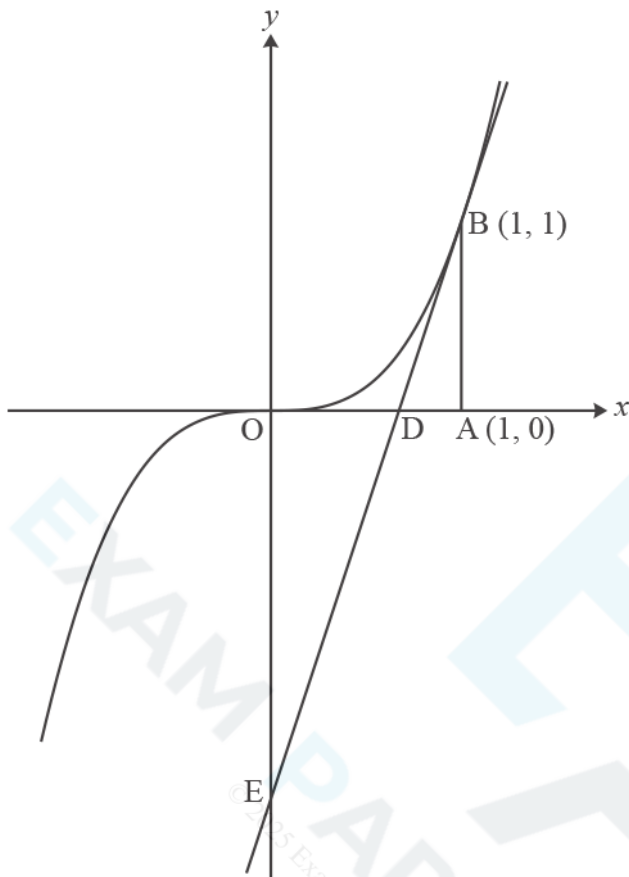


Fig. 8

Find the area of triangle ODE .

[6]

11(a) A circle has centre C (10, 4). The x-axis is a tangent to the circle, as shown in Fig. 6.

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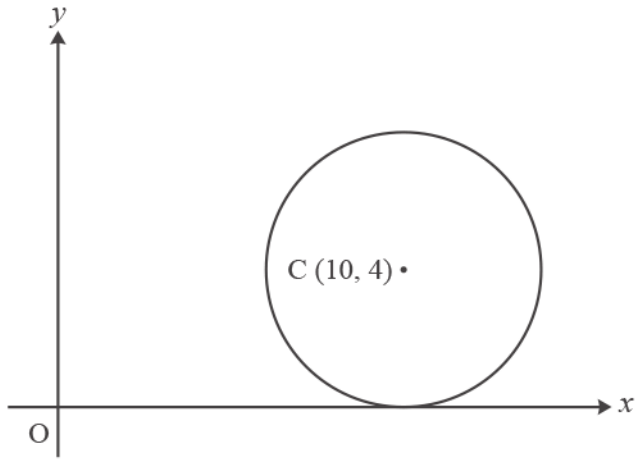


Fig. 6

Show that the line $y = x$ is not a tangent to the circle.

[4]

(b) Find the equation of the circle.

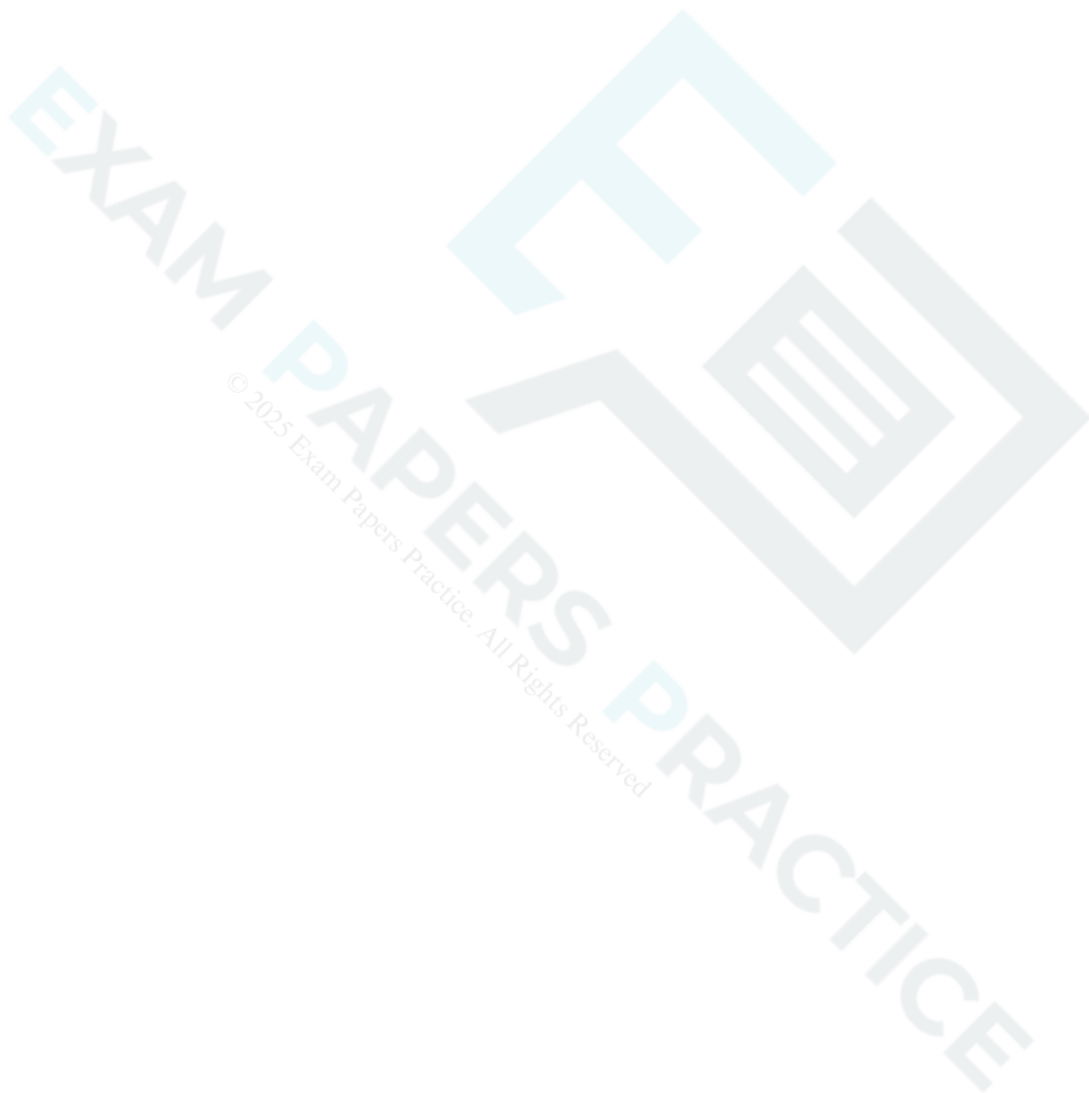
[2]

12 The function $f(x)$ is defined for all real x by

$$f(x) = 3x - 2.$$

Sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$ on the same diagram.

[2]



13(a) Fig. 12 shows a curve C with parametric equations $x = 4t^2$, $y = 8t$. The point P, with parameter t , is a general point on the curve. Q is the point on the line $x + 4 = 0$ such that PQ is parallel to the x-axis. R is the point (4, 0).

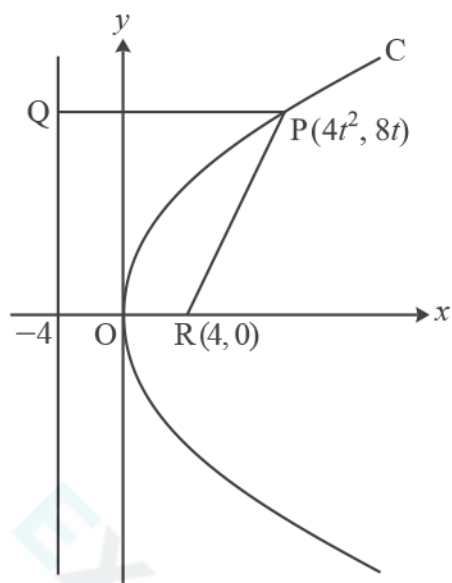


Fig. 12

Find a cartesian equation of C.

[2]

(b) Show algebraically that P is equidistant from Q and R.

[4]

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- 14 Show that the line which passes through the points $(2, -4)$ and $(-1, 5)$ does not intersect the line $3x + y = 10$.

[3]

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15 In this question you must show detailed reasoning.

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The equation of a curve is $y = \frac{x^2}{4} + \frac{2}{x} + 1$. A tangent and a normal to the curve are drawn at the point where $x = 2$.

Calculate the area bounded by the tangent, the normal and the x-axis.

[10]

16(a) The equation of a circle is $x^2 + y^2 + 8x - 6y - 39 = 0$.

Find the radius of the circle.

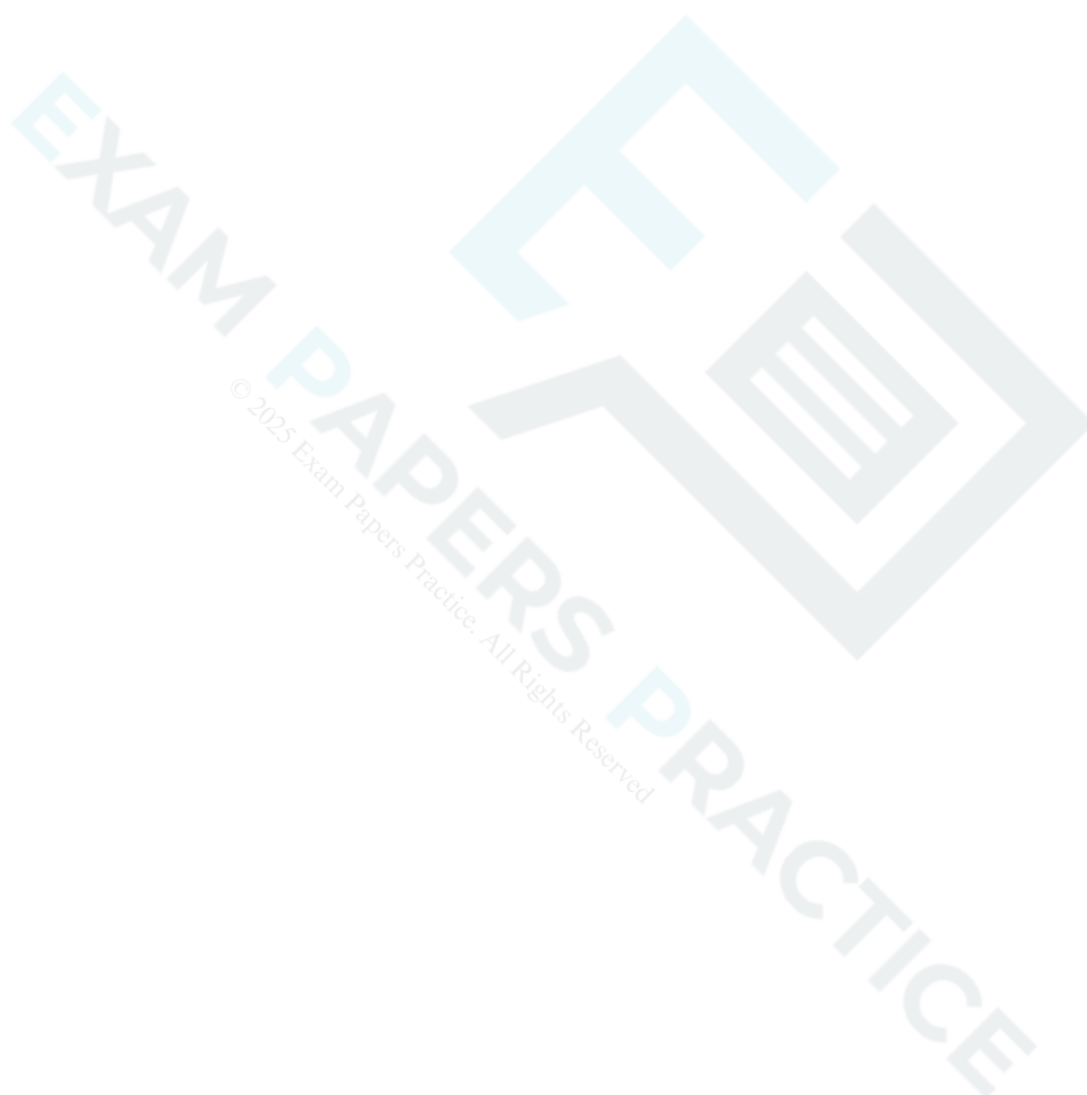
[1]

(b) Find the coordinates of the centre of the circle.



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[2]



17(a) David puts a block of ice into a cool-box. He wishes to model the mass m kg of the remaining block of ice at time t hours later. He finds that when $t = 5$, $m = 2.1$, and when $t = 50$, $m = 0.21$.

David instead proposes a linear model $m = at + b$, where a and b are constants.

Find the values of the constants for which the model fits the mass of the block when $t = 5$ and $t = 50$.

[3]

(b) Interpret these values of a and b .

[2]

(c) Find the time according to this model for the block of ice to melt completely.

[1]

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Guidance	
1	a		house prices are generally higher in London boroughs (than elsewhere in the country), so Dr Procter's suggestion is probably wrong	B1 (AO2.2a) [1]		
	b		214 505 219 402	B1 (AO3.4) B1 (AO1.1) [2]		
	c		$P = 28\,500Y - 57\,184\,000$ (where Y is the calendar year) or $P = 28\,500y + 215\,000$ (where y is the number of years after 2014)	B1 (AO3.3) B1 (AO1.1) [2]	gradient intercept	allow both marks for correct equation in any form isw allow eg $y = 28\,500x - 57\,184\,000$
	d		2016 272 000 2017 300 500	B1(AO3.4) B1(AO1.1) [2]		FT <i>their</i> straight line model provided this gives values > 250 000
	e		Dr Procter's model is a (very) poor fit Prof Jackson's is a good fit, or works well for 2017, but not 2016	B1 (AO2.2a) B1 (AO2.2a) [2]	dependent on correct values in (b) FT comment for <i>their</i> values > 250 000	this mark is dependent on having calculated values in part (d)
	f		neither – extrapolation oe	B1 (AO3.5b) [1]		
			Total	10		

Question			Answer/Indicative content	Marks	Guidance
2			Circle: equation is $(x - 2)^2 + (y + 1)^2 = 5^2$ oe Line: $m = \frac{1}{2}$ $y - 5 = (\text{their } \frac{1}{2})(x - 9)$ oe Substitution of their $y = \frac{1}{2}(x + 1)$ in their attempt at equation of circle $5x^2 - 10x - 75 = 0$ oe $x = 5$ or $x = -3$ $(5, 3)$ and $(-3, -1)$	M1 (AO1.1) A1 (AO1.1) B1 (AO1.1) M1 (AO1.1a) M1 (AO3.1a) A1 (AO1.1) A1 (AO1.1) A1 (AO2.5) [8]	Allow sign error, reversed 1 and -2, 5 not squared. Or $y - 1 = (\text{their } \frac{1}{2})(x - 1)$ Or their $x = 2y - 1$ Or $5y^2 - 10y - 15 = 0$ oe Or $y = 3$ and $y = -1$ If A0A0, sc1 for $(5, 3)$ or $(-3, -1)$ only Examiner's Comments Most candidates were able to write down the equation of the circle, though sign errors were seen both inside and between the brackets. The straight line equation was usually found correctly, and most candidates made an attempt to eliminate one variable. Although some candidates made errors in the manipulation, a good number arrived at the correct points of intersection.
			Total	8	

Question			Answer/Indicative content	Marks	Guidance
3	a	i	ke^{kx}	B1 (AO1.2) [1]	Examiner's Comments This question says 'Show that the straight line is consistent ...' so an answer should make clear how any working has shown the consistency and conclude that 'This shows the straight line is consistent with this model'.
		ii	Reason referring to growth proportional to profit.	E1 (AO3.3) [1]	E.g. As more people hear about the business, they will sell more so it is reasonable that the rate of growth is proportional to the profits
	b		$\ln y = \ln A + kx$ Equation is of form " $y = mx + c$ " so a straight line - hence model is consistent with graph	M1 (AO1.1) E1 (AO2.4) [2]	
	c		$\ln A = 1.9$ $A = 6.686$ $k = \frac{0.4}{1.6}$ $k = 0.24 \text{ to } 0.25$	M1 (AO1.1a) A1 (AO2.2a) M1 (AO1.1) A1 (AO2.2a) [4]	Intercept = 1.9 not enough for M1 One or more d.p. Attempt to find gradient Allow $e^{1.9}$ $\frac{11}{45}$ gets M1A1
	d		$x = 11$ $y = 6.686 \times e^{0.25x}$ 104.586... £104587	B1 (AO3.3) M1 (AO3.4) A1 (AO3.2a) [3]	Use of model with <i>their</i> A and k Translation into pounds (may be to nearest thousand) (FT their k and a)

Question			Answer/Indicative content	Marks	Guidance	
	e		Not reliable. Extrapolation may not be valid o.e. OR Fit of model is good so far and it's only two more years o.e.	E1 (AO3.5b) [1]	E.g. • The relationship between sales and time may change • There is a limit to the market for revision resources • Changes in the economy may affect the business	'Reliable' plus 'extrapolation' does not score
			Total	12		

Question	Answer/Indicative content	Marks	Guidance
4	EITHER Midpoint AB is (3, 1) Gradient of AB is $\frac{4 - (-2)}{7 - (-1)} = \frac{6}{8} = \frac{3}{4}$ Gradient of perpendicular $-\frac{4}{3}$ Equation of perpendicular bisector $y - 1 = -\frac{4}{3}(x - 3)$ Solve the simultaneous equations $4x + 3y = 15$ $2x + y = k$ $y = 15 - 2k, x = \frac{1}{2}(3k - 15)$ giving P $\left(\frac{1}{2}(3k - 15), 15 - 2k\right)$ OR P is of the form (x, k - 2x) and is equidistant from A and B $(x + 1)^2 + (y + 2)^2 = (x - 7)^2 + (y - 4)^2$ $(x + 1)^2 + (k - 2x + 2)^2 = (x - 7)^2 + (k - 2x - 4)^2$ $-8x + 1 = 60 - 12k$ So $x = \frac{1}{2}(3k - 15)$ and $y = 15 - 2k$ P is $\left(\frac{1}{2}(3k - 15), 15 - 2k\right)$	B1(AO3.1a) M1(AO3.1a) M1(AO1.1a) M1(AO1.1a) M1(AO1.1a) M1(AO1.1a) A1(AO1.1) A1(AO1.1) [7] B1 M1 M1 M1 M1 A1 A1 [7]	Seen or implied Must be correct way up – allow one slip negative reciprocal FT their gradient of AB FT their gradient. Condone A or B used instead of M. Do not allow if their gradient of AB used. Attempt to eliminate one variable cao cao Seen or implied Finding at least one distance Equating distances Substituting for y Attempt to simplify cao cao Allow this M1 for showing the perpendicular to the given line is

Question			Answer/Indicative content	Marks	Guidance
			Total	7	
5			$u_x = 35 \cos \theta$ giving $x = (35 \cos \theta)t$ $u_y = 35 \sin \theta$ giving $y = (35 \sin \theta) t - \frac{1}{2}gt^2$ Substitute for t $y = (35 \sin \theta) \left(\frac{x}{35 \cos \theta} \right) - \frac{1}{2}g \left(\frac{x}{35 \cos \theta} \right)^2$ $\left[y = x \tan \theta - \frac{x^2}{250 \cos^2 \theta} \right]$	B1(AO3.3) B1(AO3.3) M1(AO3.3) A1(AO1.1) [4]	Award seen in any form soi Substituting for t in their equation for y Award in any form ISW
			Total	4	

Question		Answer/Indicative content	Marks	Guidance	
6	a	DR $\frac{dx}{dt} = -2t^{-3} \text{ and } \frac{dy}{dt} = -3t^{-4} + t^{-2}$ $\text{So } \frac{dy}{dx} = \frac{-3t^{-4} + t^{-2}}{-2t^{-3}}$ Multiply top and bottom by t^4 $\frac{dy}{dx} = \frac{-3 + t^2}{-2t} = \frac{3 - t^2}{2t}$	M1(AO2.1) M1(AO2.1) A1(AO2.1) [3]	Attempt to differentiate both equations Combining derivatives for $\frac{dy}{dx}$ AG Correct derivative in required form.	Note that $\frac{dy}{dx} = \left(-\frac{3}{t^4} - \frac{1}{t^2}\right) \times \left(-\frac{t^3}{2}\right)$
	b	DR tangent parallel when $\frac{dy}{dx} = -\frac{1}{4}$ $\frac{3 - t^2}{2t} = -\frac{1}{4}$ $4t^2 - 2t - 12 = 0$ roots $2, \left[-\frac{3}{2}\right]$ [but since $t > 0$ $t = 2$] When $t = 2$, $x = \frac{1}{4}$, $y = \frac{1}{8} - \frac{1}{2} = -\frac{3}{8}$ So the coordinates are $\left(\frac{1}{4}, -\frac{3}{8}\right)$	B1(AO3.1a) M1(AO1.1a) A1(AO1.1) [3]	Establishing gradient $-\frac{1}{4}$ Forming and solving quadratic equation. Using the value of t for both coordinates	$y = -\frac{1}{4}x + \frac{1}{4}$ not sufficient on its own Ignore any point based on $t = -\frac{3}{2}$

Question		Answer/Indicative content	Marks	Guidance
	c	DR Rearrange $t = x^{-\frac{1}{2}}$ Substitute $y = \left(x^{-\frac{1}{2}}\right)^{-3} - \left(x^{-\frac{1}{2}}\right)^{-1} = x^{\frac{3}{2}} - x^{\frac{1}{2}}$ $= x^{\frac{1}{2}}(x-1) = (x-1)\sqrt{x}$	B1(AO3.1a) M1(AO1.1) A1(AO1.1) [3]	or equivalent eg $\frac{1}{t} = \sqrt{x}$ Attempt to eliminate t factorized form. Allow surd or index form $y = \frac{1}{\left(\frac{1}{\sqrt{x}}\right)^3} - \frac{1}{\left(\frac{1}{\sqrt{x}}\right)}$ Do not allow for $= \pm(x-1)\sqrt{x}$
		Total	9	

Question			Answer/Indicative content	Marks	Guidance
7	a		$m = \frac{3.58-3.93}{4-0}$ $\log_{10} V = -0.0875t + 3.93$	M1 (AO1.1) M1 (AO3.3) A1 (AO1.1) [3]	(±) 0.0875 oe Allow any value for – 0.0875 condone omission of base Condone ln Examiner's Comments Only about half the candidates realised that they needed to find the gradient and intercept of the line in Fig. 9.2 and then form an equation for $\log_{10} V$ in terms of t. Some of these candidates made errors in finding the gradient of the line.
	b		$V = 10^{-0.0875t + 3.93}$ $A = 10^{3.93} = 8500$ $b = 10^{-0.0875} = 0.82$	M1 (AO3.1b) A1 (AO3.3) A1 (AO1.1) [3]	Or $\log V = \log A + \log b^t$ 8511.38 Accept $V = 8500 \times 0.82^t$ 0.81752; sc B1 if both answers not to 2 sf. Allow 8500 taken from table (M1A1) Examiner's Comments Most candidates were able to find the value of A, but many had difficulty finding the value of b. Some candidates failed to give the answers correct to 2 significant figures.

Question			Answer/Indicative content	Marks	Guidance
	c		3151.29 (~ 3150) so the model is a (very) good fit	E1 (AO3.4) [1]	Must see value from model OR find value of t (approx. 4.98) when model gives 3150 Eg 3107.93 from more accurate values. <u>Examiner's Comments</u> This part was omitted by a considerable number of candidates. The command word 'determine' is intended to convey to candidates that a calculation may well be needed to justify their answer. Answers without both an approximately correct value and a correct conclusion were given no credit. Candidates were expected to show that the model in part (b) gives a value for the car of approximately £3151 after 5 years, and to conclude that the model was a (very) good fit. Answers were also accepted where candidates used more accurate values of A and b in the model, where candidates used the expression in part (a) to work out the value of the car after 5 years, and where candidates showed that the model would give the value of £3150 after about 4.98 years, which is very close to 5.

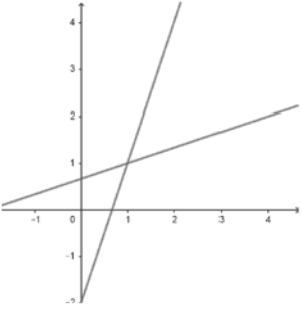
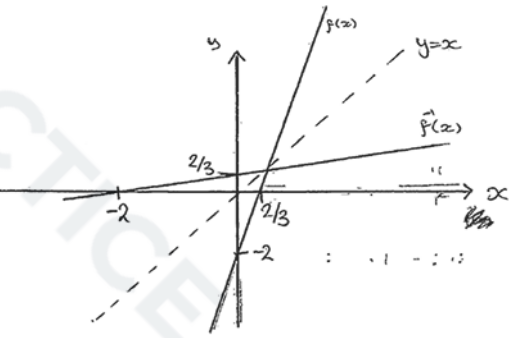
$$t = \frac{h}{r}$$

Question	Answer/Indicative content	Marks	Guidance
9	DR Gradient of normal = 2 $\Rightarrow$ gradient of tangent = -0.5 $2y \frac{dy}{dx} + y + x \frac{dy}{dx} = 0$ eg $-y + y - 0.5x = 0 \Rightarrow x = 0$ $y^2 = 8$ $(0, \sqrt{8})$ and $(0, -\sqrt{8})$ Alternative method $\frac{dx}{dy} = \frac{-8 - y^2}{y^2} \text{ oe}$ $\frac{8 + y^2}{y^2} = 2 \text{ oe}$ $y^2 = 8$ $(0, \sqrt{8})$ and $(0, -\sqrt{8})$	M1 (AO3.1a) M1 (AO1.1) A1 (AO1.1) A1 (AO2.2a) M1 (AO1.1) A1 (AO1.1) M1 M1 A1 M1 M1 A1 [6]	$\frac{-y}{x+2y} = -\frac{1}{2}$ Attempt at implicit differentiation with one algebraic term differentiated correctly Correct implicit expression Implied by at least one y-value Or M1 for correct expression for x in terms of y and attempt to differentiate For one step solving their equation in x and/or y with original equation Rearranging to $x = \frac{8 - y^2}{y}$ and attempting differentiation Differentiation by dividing first or quotient rule – allow one error Correct differentiation Using negative reciprocal Implied by at least one y-value Examiner's Comments There seemed to be a good understanding of the need for implicit differentiation, and many fully correct solutions were seen. The marking of this question allowed for those with the correct concepts, but poor algebraic manipulative skills to achieve a

Question			Answer/Indicative content	Marks	Guidance	
					reasonable number of marks. A few candidate tried to rearrange to get a function of the form $y = g(x)$, but this then lead to greater difficulty in completing it correctly. Many got through the whole question but did not get the last mark for only giving one solution to $y^2 = 8$.	
			Total	6		
10			DR $\frac{dy}{dx} = 3x^2$ Gradient at (1, 1) is 3 Tangent is $(y - 1) = 3(x - 1)$ OE = 2 and OD = $\frac{2}{3}$ Area triangle ODE = $\frac{1}{2} \times \frac{2}{3} \times 2 = \frac{2}{3}$	M1 (AO3.1a) M1 (AO1.1) A1 (AO1.1) M1 (AO3.1a) A1 (AO1.1) A1 (AO1.1) [6]	$(y = 3x - 2;$ simplified form not essential) Finding either of these lengths soi by E (0, -2) or $D\left(\frac{2}{3}, 0\right)$ Bothe OE and OD correct	or $\frac{AB}{AD} = 3$ or $AD = \frac{1}{3}$ Or Triangle ODE is similar to ABD [with scale factor 2] Examiner's Comments This question was very well answered with many students gaining full marks. They clearly understood the aim of the question and the methods needed to solve. The majority of errors seen were simple slips in arithmetic. The great majority of candidates realised that they needed to find the equation of the straight line in order to find the lengths of OE and OD. Those who differentiated the equation of the curve correctly were generally successful in the whole question.
			Total	6		

Question		Answer/Indicative content	Marks	Guidance	
11	a	Where $y = x$ meets the circle $(x - 10)^2 + (x - 4)^2 = 16$ $2x^2 - 28x + 116 = 16$ or $x^2 - 14x + 50 = 0$ or $2x^2 - 28x + 100 = 0$ $b^2 - 4ac = -4$ or -16 so no meeting points i.e. not a tangent Alternative method Angle between $y = x$ and x-axis = 45° Let θ be the angle between the x-axis and the line joining $(0, 0)$ and $(10, 4)$. $\tan \theta = 0.4$ $\theta = 21.80(2 \text{ d.p.})$ If $y = x$ was a tangent θ would be 22.5° hence not a tangent	M1 (AO3.1a) M1 (AO1.1) M1 (AO1.1) E1 (AO2.4) M1 M1 M1 E1	For sub'n of $y = x$ into their circle equation For expanding and collecting like terms Rearranging to 3 term quadratic = 0 Or $(x - 7)^2 + 1 = 0$ no meeting points hence not a tangent Alternative method 2 gradient of normal is -1 Line joining point of contact (k, k) to centre $(10, 4)$ must have gradient -1 $k = 7$ $(7, 7)$ does not lie on circle	'No real roots' not enough for E1 Allow E1 for correct reasoning following omission of '=0'
		Examiner's Comments The common method for this part was to solve their equation from the part above simultaneously with $y = x$ and the method was generally competently shown the difficulty coming with finishing the question off with many leaving their answer as discriminant < 0, or 'no real roots' or giving the complex roots and not completing by stating 'hence it is not a tangent'.			

Question			Answer/Indicative content	Marks	Guidance
	b		Radius = 4 $(x - 10)^2 + (y - 4)^2 = 16$	B1 (AO3.1a) B1 (AO1.1) [2]	soi by 16 o.e. <u>Examiner's Comments</u> Nearly all candidates earned both marks in this part with only a few using other than 4 for the radius (2 and 10 were both seen) and a few candidates who missed the squares or did minus between the brackets or no sign at all.
			Total	6	

Question	Answer/Indicative content	Marks	Guidance
12		B1 (AO1.1) B1 (AO1.1) [2]	$y = 3x - 2$ going through $(0, -2)$ Or $y = \frac{x+2}{3}$ going through $(0, 2/3)$ $y = f^{-1}(x)$ a reflection of $y = f(x)$ in $y = x$ Both graphs linear to get B2 Line of symmetry roughly at 45° or implied by coords Examiner's Comments Most candidates provided good quality sketches and understood the need for the inverse to be a reflection in $y = x$; the best solutions added $y = x$ to their sketches. While a sketch does not need scaled axes, 'important points' should be marked such as the intercepts. Exemplar 1  This candidate clearly shows the line of symmetry making it easier for them to draw the graphs and the examiner see their intent. They clearly indicate the y-

Question			Answer/Indicative content	Marks	Guidance
					intercepts and x-intercepts as per the notes in the specification about the level of detail required for a sketch.
			Total	2	

EXAM PAPERS PRACTICE
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Question			Answer/Indicative content	Marks	Guidance
13	a		Substitute $t = \frac{y}{8}$ to obtain $x = 4\left(\frac{y}{8}\right)^2$ $y^2 = 16x$	M1 (AO 1.1a) A1 (AO 1.1b) [2]	Rearranged equation must be used Allow any equivalent form including $y = \pm 4\sqrt{x}$ allow use of $t = \frac{\sqrt{x}}{2}$ oe used for M1A0. Examiner's Comments This was generally well done, with most candidates using the parametric equations from the stem of the question without comment.

Question		Answer/Indicative content	Marks	Guidance
	b	$x = 4t^2, y = 8t$ $PQ = 4 + 4t^2$ $PR^2 = (4t^2 - 4)^2 + (8t)^2$ $= 16t^4 + 32t^2 + 16$ $= (4 + 4t^2)^2 = PQ^2$ So equidistant	B1 (AO 2.1) M1 (AO 2.1) M1 (AO 2.1) A1 (AO 2.1) [4]	Use of distance formula Must show some algebraic manipulation Correct expression for PR or PR^2 Conclusion comparing PQ and PR or PQ^2 and PR^2 Examiner's Comments We are sorry for the error on this question. During marking, we looked at the performance of candidates taking the paper. As a result, we decided the fairest approach was to award all candidates full marks for question 12(a) and question 12 (b). However, we are pleased that there was little evidence that this question had an impact on candidate progress on the rest of the paper. The majority of candidates took the information for their calculations from the diagram and appeared not to notice that there was an error in the second equation in the stem of the question. Only a small minority made any comment about the error or used both given expressions for y . The algebraic manipulation required in this question is quite tricky and many good solutions were seen, as were very poor or fragmented solutions. A very small number of candidates who had used the value from the stem then repeated the work with the value from the diagram and were thus disadvantaged in the examination.
		Total	6	

Question			Answer/Indicative content	Marks	Guidance	
14			(Gradient of line segment) $= \frac{5 - (-4)}{-1 - 2} = -3$ (Given line is) $y = -3x + 10$ has gradient -3 Same gradient so parallel lines Neither point lies on the line so the lines do not intersect Alternative solution $\text{Gradient} = \frac{5 - (-4)}{-1 - 2} = -3$ Equation of the line $y - (-4) = -3(x - 2)$ $y = -3x + 2$ Given line is $y = -3x + 10$ which is parallel [So the lines do not intersect]	M1 (AO 3.1a) M1 (AO 1.1a) A1 (AO 2.2a) [3] M1 M1 A1	Attempt to find gradient –accept sign errors but not reciprocal Finding gradient of given line Must make conclusion based on the fact that the two lines are parallel and not the same line Attempt to find gradient –accept sign errors but not reciprocal Finding equation of given line Conclusion referring to parallel lines	Allow for solving two lines simultaneously and stating there are no solutions. Examiner's Comments Most candidates correctly found the equation of the line joining the points. When solving the two equations simultaneously, some would stop at $2 \neq 10$ with no further comment. Many candidates who found only gradients commented that the lines were parallel but did not consider the possibility that the given points were on the line $3x + y = 10$. Fortunately, relatively few thought that it was sufficient to show that either point lies on the given line (this received no marks).
			Total	3		

Question			Answer/Indicative content	Marks	Guidance
15			$x = 2, y = 3$ $\frac{dy}{dx} = \frac{2x}{4} - \frac{2}{x^2}$ oe Evaluaton of their derivative at $x = 2$ $y - 3 = \frac{1}{2}(x - 2)$ $y - 3 = -2(x - 2)$ intercepts are $(-4, 0)$ and $(3.5, 0)$ $\text{Area} = \frac{1}{2} \times 7 \frac{1}{2} \times 3$ $\frac{45}{4}$ cao	B1 (AO1.1) M1 (AO2.1) A1 (AO1.1) M1 (AO1.1) M1 (AO2.4) M1 (AO1.1) A1 (AO1.1) A1 (AO1.1) M1 (AO2.1) A1 (AO1.1) [10]	soi Differentiation with 1 term correct NB $\frac{1}{2}$ equation of tangent $y = \frac{1}{2}x + 2$ $y = -2 + 7$ and equation of normal; FT their tangent gradient OR $\frac{1}{2} \times \sqrt{(6^2 + 3^2)} \times \sqrt{(3^2 + 1.5^2)}$ - lengths of tangent and normal or 11.25 <u>Examiner's Comments</u> This question was found accessible by almost all candidates, with a considerable number giving completely correct answers. However a small minority of candidates who integrated the equation of the curve – presumably triggered by the word ‘area’. Candidates started by differentiating the equation of the curve. Candidates making errors in this step were still able to gain marks for attempting to find the slope of the tangent when $x = 2$, attempting to find the equations of the tangent and normal, and for attempting to find the area between the tangent, normal and the x-axis. They were also able to gain a mark for finding the value of y when $x = 2$. It was noted that candidates drawing a sketch of the tangent, normal and x-axis were more successful in identifying the area required and to calculate it efficiently and correctly.
			Total	10	

Question			Answer/Indicative content	Marks	Guidance
16	a		radius is 8	B1 (AO1.1) [1]	<u>Examiner's Comments</u> Candidates with the centre wrong in part (a) usually got this part wrong, as did those whose rearrangement of the circle equation in part (a) went astray when finding the constant term.
	b		$(x \pm 4)^2$ and $(y \pm 3)^2$ seen centre is $(-4, 3)$	M1 (AO3.1a) A1 (AO1.1) [2]	$(x + 4)^2 - 4^2 +$ $(y - 3)^2 - (-3)^2 - 39$ $= 0$ www <u>Examiner's Comments</u> Most candidates were able to complete the square correctly and give the coordinates of the centre, though a few made sign slips. A significant number of candidates did not recognise the need to complete the square, writing things like $x(x + 8) + y(y - 6)$ and often giving the answer as $(-8, 6)$.
			Total	3	

Question		Answer/Indicative content	Marks	Guidance	
17	a	Substitute $t = 5$, $m = 2.1 \Rightarrow 2.1 = 5a + b$ Substitute $t = 50$, $m = 0.21 \Rightarrow 0.21 = 50a + b$ Solving simultaneously: $a = -0.042$, $b = 2.31$ Alternative method gradient of line joining points (5,2.1) and (50,0.21) $\frac{2.1 - 0.21}{5 - 50} = -0.042$ So line is of the form $y - 2.1 = -0.042(x - 5)$ $y = -0.042x + 2.31$	M1 (AO3.3) M1 (AO1.1a) A1 (AO1.1) [3] M1 M1 A1 [3]	Uses either data point to form equation Uses the other data point to find 2nd equation and solve simultaneously Both values required Attempt to find gradient Uses either data point to complete the equation Award for the correct equation without reference to a and b	Solution may be BC

Examiner's Comments

Most candidates substituted both pairs of values and solved the simultaneous equations and were largely successful.



AfL

This is exactly the sort of question where using the simultaneous equations facility of the calculator is appropriate and expected. There is no need for detailed reasoning, the coefficients are not easy to work with and the question is only worth 3 marks including the marks for forming the equations.

Question			Answer/Indicative content	Marks	Guidance	
	b		a is the rate at which ice melts: 0.042 kg of ice is lost per hour b is the initial mass of the block: 2.31 kg	B1 (AO3.3) B1 (AO3.3) [2]	Must refer to the value found; FT their negative a FT their positive b	If no reference is made to the numerical values, allow SC1 for ' a is the rate at which ice melts and b is the initial mass of the block' oe
					Examiner's Comments For full credit the candidates needed to refer to the specific values and what they meant in the context of the physical situation.	
	c		$m = 0 \Rightarrow 0 = -0.042t + 2.31 \Rightarrow$ $t = \frac{2.31}{0.042} = 55$ so time for block to melt is 55 hours	B1 (AO3.4) [1]	FT their a and b only if t is positive	Examiner's Comments Most candidates who had solved the simultaneous equations in part (c) were able to get this mark as follow through was allowed. Candidates who did not have a negative value for a were not awarded the mark for conveniently not noticing that their t should have been negative.
			Total	6		